

CDC Diabetes Health Indicators Analysis

Group Project

2025-12-03

Introduction

This project analyzes the CDC Diabetes Health Indicators dataset to predict whether an individual has diabetes based on various lifestyle factors and health indicators. The goal is to build a binary classification model that can accurately identify individuals at risk of diabetes. We will explore the data, handle multicollinearity, and compare different logistic regression models, including those with regularization (LASSO and Ridge).

Data Description

The dataset is obtained from the UCI Machine Learning Repository (ID: 891). It contains health indicators for 253,680 individuals.

Raw Data and Pre-processing

The raw data was fetched using the `ucimlrepo` Python package and exported to a CSV file. The dataset includes a binary target variable `Diabetes_binary` (0 = no diabetes, 1 = pre-diabetes or diabetes) and 21 feature variables.

The dataset contains 253680 observations and 22 variables. There are 0 missing values in the dataset.

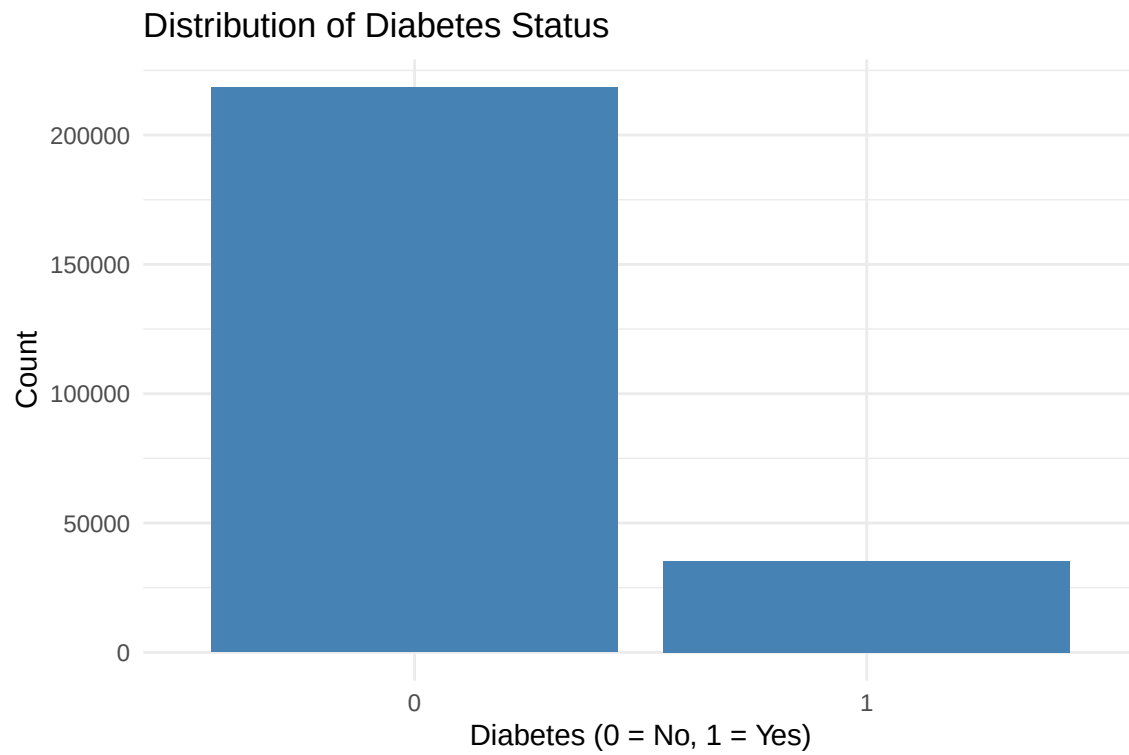
Predictor Selection

We initially consider all available predictors. However, we will check for multicollinearity and use LASSO regularization to select the most relevant variables.

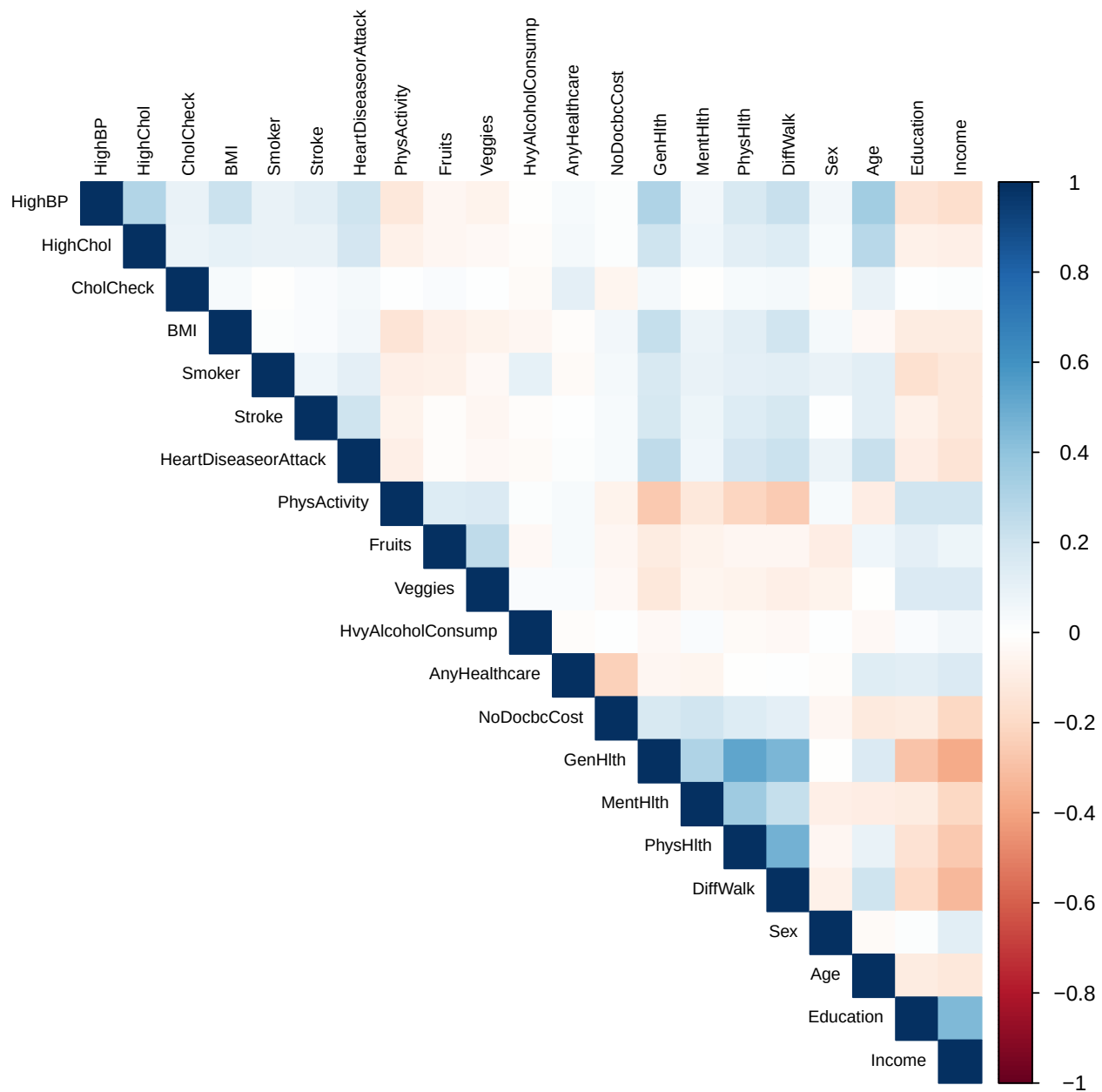
The predictors include: `HighChol`, `CholCheck`, `BMI`, `Smoker`, `Stroke`, `HeartDiseaseorAttack`, `PhysActivity`, `Fruits`, `Veggies`, `HvyAlcoholConsump`, `AnyHealthcare`, `NoDocbcCost`, `GenHlth`, `MentHlth`, `PhysHlth`, `DiffWalk`, `Sex`, `Age`, `Education`, `Income`, `Diabetes_binary`

Exploratory Data Analysis (EDA)

Let's look at the distribution of the target variable.



We can also examine the correlation between numeric predictors to identify potential multicollinearity.



Methodology

We will split the data into training (80%) and testing (20%) sets.

Logistic Regression

We fit a standard logistic regression model using all predictors.

```
##
## Call:
## glm(formula = Diabetes_binary ~ ., family = "binomial", data = dataTrain)
```

```
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -7.9387392   0.1062710 -74.703 < 2e-16 ***
## HighBP          0.7395755   0.0165125  44.789 < 2e-16 ***
## HighChol        0.5774981   0.0151904  38.017 < 2e-16 ***
## CholCheck       1.3056708   0.0786529  16.600 < 2e-16 ***
## BMI             0.0616974   0.0010131  60.897 < 2e-16 ***
## Smoker          -0.0027705   0.0147521  -0.188 0.851030
## Stroke           0.1455569   0.0279613   5.206 1.93e-07 ***
## HeartDiseaseorAttack 0.2304988   0.0198750  11.597 < 2e-16 ***
## PhysActivity     -0.0590366   0.0161459  -3.656 0.000256 ***
## Fruits          -0.0486044   0.0153001  -3.177 0.001489 **
## Veggies         -0.0405459   0.0177837  -2.280 0.022611 *
## HvyAlcoholConsump -0.7691523   0.0429250 -17.919 < 2e-16 ***
## AnyHealthcare     0.0753786   0.0374839   2.011 0.044330 *
## NoDocbcCost       0.0278689   0.0257776   1.081 0.279641
## GenHlth          0.5358064   0.0091009  58.874 < 2e-16 ***
## MentHlth         -0.0040714   0.0009576  -4.251 2.12e-05 ***
## PhysHlth         -0.0080602   0.0008765  -9.196 < 2e-16 ***
## DiffWalk         0.1353457   0.0189731   7.134 9.78e-13 ***
## Sex              0.2551036   0.0150420  16.959 < 2e-16 ***
## Age              0.1253120   0.0031349  39.974 < 2e-16 ***
## Education        -0.0275403   0.0078087  -3.527 0.000420 ***
## Income          -0.0492259   0.0040012 -12.303 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 163879  on 202944  degrees of freedom
## Residual deviance: 129755  on 202923  degrees of freedom
## AIC: 129799
##
## Number of Fisher Scoring iterations: 6
```

Regularization (LASSO and Ridge)

We use `glmnet` to fit LASSO ($\alpha = 1$) and Ridge ($\alpha = 0$) models. We use cross-validation to find the optimal λ .

LASSO (L1 Regularization)

```
## 22 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## (Intercept)    -7.790869077
## HighBP          0.737268295
## HighChol        0.571313211
## CholCheck       1.213047392
## BMI             0.061135427
## Smoker          .
## Stroke           0.137848558
## HeartDiseaseorAttack 0.229972905
```

## PhysActivity	-0.053535556
## Fruits	-0.042212614
## Veggies	-0.036216707
## HvyAlcoholConsump	-0.735649764
## AnyHealthcare	0.047625368
## NoDocbcCost	0.006377872
## GenHlth	0.530017898
## MentHlth	-0.003562326
## PhysHlth	-0.007168251
## DiffWalk	0.130093219
## Sex	0.243940698
## Age	0.123762146
## Education	-0.026107096
## Income	-0.047962741

The optimal lambda for LASSO is 4.1903019×10^{-4} . LASSO performs variable selection by shrinking some coefficients to zero.

Ridge (L2 Regularization)

The optimal lambda for Ridge is 0.0101411.

Results

We compare the models based on accuracy on the test set.

- **Logistic Regression Accuracy:** 0.8643
- **LASSO Accuracy:** 0.8644
- **Ridge Accuracy:** 0.8642

Conclusion

In this analysis, we built logistic regression models to predict diabetes. We compared a standard model with LASSO and Ridge regularization.

(Note: Specific conclusions will depend on the actual run results).

Appendix: R Code

```
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)
library(tidyverse)
library(glmnet)
library(caret)
library(corrplot)
# Load data
data <- read.csv("diabetes_data.csv")

# Convert target to factor
data$Diabetes_binary <- as.factor(data$Diabetes_binary)

# Check for missing values
missing_values <- sum(is.na(data))
ggplot(data, aes(x = Diabetes_binary)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Distribution of Diabetes Status", x = "Diabetes (0 = No, 1 = Yes)", y = "Count") +
  theme_minimal()

# Calculate correlation matrix for numeric predictors
numeric_vars <- data %>% select(-Diabetes_binary)
cor_matrix <- cor(numeric_vars)
corrplot(cor_matrix, method = "color", type = "upper", tl.col = "black", tl.cex = 0.7)
set.seed(123)
trainIndex <- createDataPartition(data$Diabetes_binary, p = .8,
                                   list = FALSE,
                                   times = 1)

dataTrain <- data[ trainIndex,]
dataTest <- data[-trainIndex,]
model_glm <- glm(Diabetes_binary ~ ., data = dataTrain, family = "binomial")
summary(model_glm)
x_train <- model.matrix(Diabetes_binary ~ ., dataTrain)[,-1]
y_train <- dataTrain$Diabetes_binary

x_test <- model.matrix(Diabetes_binary ~ ., dataTest)[,-1]
y_test <- dataTest$Diabetes_binary
cv_lasso <- cv.glmnet(x_train, y_train, alpha = 1, family = "binomial")
best_lambda_lasso <- cv_lasso$lambda.min
model_lasso <- glmnet(x_train, y_train, alpha = 1, family = "binomial", lambda = best_lambda_lasso)

# Coefficients
coef(model_lasso)
cv_ridge <- cv.glmnet(x_train, y_train, alpha = 0, family = "binomial")
best_lambda_ridge <- cv_ridge$lambda.min
model_ridge <- glmnet(x_train, y_train, alpha = 0, family = "binomial", lambda = best_lambda_ridge)

# Logistic Regression Predictions
prob_glm <- predict(model_glm, dataTest, type = "response")
pred_glm <- ifelse(prob_glm > 0.5, 1, 0)
acc_glm <- mean(pred_glm == dataTest$Diabetes_binary)

# LASSO Predictions
prob_lasso <- predict(model_lasso, newx = x_test, type = "response")
pred_lasso <- ifelse(prob_lasso > 0.5, 1, 0)
```

```
acc_lasso <- mean(pred_lasso == y_test)

# Ridge Predictions
prob_ridge <- predict(model_ridge, newx = x_test, type = "response")
pred_ridge <- ifelse(prob_ridge > 0.5, 1, 0)
acc_ridge <- mean(pred_ridge == y_test)
```